EXP.NO:23

# Print Intersection of 2 sorted arrays-(m+n) Time Complexity, O(1) Space Complexity

**PROBLEM STATEMENT:**

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

- The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

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**CODE**

```c
#include <stdio.h>
#define MAX 10000
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    int seen[MAX]={0};
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n;i++){
        seen[a[i]]++;
    }
    for(int i=0;i<n;i++){
        if(seen[a[i]]>1){
            printf("%d",a[i]);
            return 0;
        }
    }
}
```

**OUTPUT:**

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 | 10 57 | ✔ |
| ✔ | 1<br>6 1 2 3 4 5 6<br>2<br>1 6 | 1 6 | 1 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

**RESULT:**

The code is executed successfully.